

Euclid, Book I, Proposition I.6

If Two Angles of a Triangle Are Equal, Then the Opposite Sides Are Equal

CGJteamLab

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1 Introduction

Proposition I.6 establishes the converse of the preceding proposition. Where Proposition I.5 proves that equal sides of a triangle imply equal base angles, Proposition I.6 shows that equal base angles imply the equality of the opposite sides.

This result completes the fundamental correspondence between the sides and angles of an isosceles triangle. Together, Propositions I.5 and I.6 establish the equivalence between the two defining properties of an isosceles triangle.

Euclid's original proof is considerably more involved than the modern reconstruction. It proceeds by contradiction, introducing an auxiliary construction obtained from Proposition I.3 and ultimately deriving an impossible configuration by repeated use of Proposition I.4.

In the Hilbert reconstruction developed in this project, the theorem admits a substantially shorter proof. The previously established Angle–Side–Angle theorem is sufficient to derive the equality of the opposite sides directly, eliminating the auxiliary construction and the *reductio ad absurdum* argument required by Euclid.

2 The Classical Configuration

Euclid's proof begins with a triangle whose base angles are assumed to be equal. The desired conclusion is that the opposite sides are also equal.

Instead of arguing directly, Euclid assumes that the two sides are unequal and derives a contradiction. The proof therefore combines an auxiliary construction with a *reductio ad absurdum* argument.

The following sequence illustrates the successive stages of Euclid's construction.

2.1 Initial Triangle

Let ABC be a triangle satisfying

$$\angle ABC \cong \angle ACB.$$

At this stage no auxiliary construction has yet been introduced.

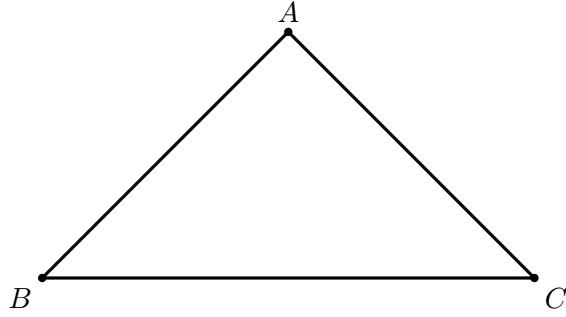


Figure 1: The data of Proposition I.6: the base angles satisfy $\angle ABC \cong \angle ACB$.

2.2 Assuming Unequal Sides

To obtain a contradiction, Euclid assumes that the two sides opposite the equal angles are not equal. Without loss of generality, suppose

$$AB > AC.$$

Using Proposition I.3, a point D is chosen on the segment AB so that

$$DB \cong AC.$$

Consequently,

$$A - D - B.$$

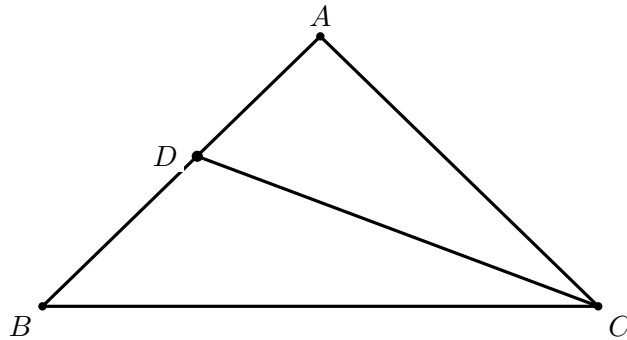


Figure 2: Euclid's contradiction configuration for the case $AB > AC$. Proposition I.3 chooses D on AB with $DB \cong AC$, and D is joined to C . The congruence is stated in the text rather than marked again on the diagram.

3 Statement

Theorem 1 (Euclid I.6). *Let ABC be a nondegenerate triangle.*

If

$$\angle ABC \cong \angle ACB,$$

then

$$AB \cong AC.$$

That is, equal angles in a triangle subtend equal opposite sides.

Diagrammatic audit. The three former auxiliary pictures were successive snapshots of the same construction and have been merged into one. Two figures are sufficient: the initial triangle and the contradiction configuration with the point D and the segment DC .

Wilkins’s discussion confirms that the essential classical move is exactly this one: assuming, say, $AB > AC$, choose $D \in AB$ with $DB = AC$, then compare DBC with ABC by SAS. His note also records an alternative angle-based contradiction, but that is another proof rather than another necessary diagram for the reconstruction.

4 Classical Proof

Euclid proves the theorem by contradiction.

Assume that the sides opposite the equal angles are unequal. Without loss of generality, suppose that AB is greater than AC . By Proposition I.3, a point is constructed on the longer side so that the remaining segment is equal to the shorter side.

Joining the auxiliary point to the remaining vertex produces a second triangle. Proposition I.4 is then applied to show that the auxiliary triangle is congruent to the original one.

The resulting congruence forces equalities that are incompatible with the assumption that one side is strictly longer than the other. Since the assumption leads to a contradiction, the two sides must be equal.

Thus equal angles in a triangle necessarily subtend equal opposite sides.

5 Proof Architecture of the Reconstruction

The formal proof follows a substantially different route from Euclid’s classical argument. Rather than introducing an auxiliary point, constructing additional triangles, and reasoning by contradiction, the implementation exploits a higher-level theorem already available in the Hilbert interface.

The proof compares the given triangle with itself after exchanging the roles of the two equal angles. Since the common side is trivially congruent to itself, the derived Angle–Side–Angle theorem immediately implies the equality of the remaining sides.

Consequently, the entire classical construction disappears from the formal proof and is replaced by a single application of a previously established congruence theorem.

Euclid

```

-----
ABC with angle ABC ~= angle ACB
  |
assume AB != AC
  |
choose one strict inequality, e.g. AB > AC
  |
I.3: choose D on AB with DB ~= AC
  |
join DC
  |
I.4: SAS
  |
auxiliary triangle equals whole triangle
  |
contradiction
  |
AB ~= AC

```

Hilbert / Lean

```

-----
angle ABC ~= angle ACB
  |
reorient the two angle congruences
  |
BC ~= CB
  |
hilbert_asa_sides
on triangles BCA and CBA
  |
AB ~= AC

```

6 Hilbert Reconstruction

The formal proof uses a substantially shorter route than Euclid's contradiction argument. It applies an established ASA theorem directly.

From the hypothesis

$$\angle ABC \cong \angle ACB$$

the proof first obtains the corresponding reversed angle congruences needed for the triangles

$$\triangle BCA \quad \text{and} \quad \triangle CBA.$$

The side BC is common to both triangles, so

$$BC \cong CB.$$

The theorem `hilbert_asa_sides` then yields the equality of the remaining corresponding sides. After reorienting the segment congruence, this is exactly

$$AB \cong AC.$$

Thus no segment trichotomy, point D , Proposition I.3 construction, or contradiction appears in `Proposition06.lean`. Those belong to the classical Euclidean proof, not to the formal wrapper used here.

7 Lean Formalization

The complete implementation is available in the repository. Only the two steps that explain the formal proof are reproduced here.

First the common side and the two angle orientations are prepared. The decisive theorem call is then:

```
have hASA :=
  hilbert_asa_sides
    Geo
    B C A
    C B A
    hBCA
    hCBA
    hBC
    hAngleB
    hAngleC
```

The returned side congruence has the orientation dictated by those two triangles, so the final line merely reverses its endpoints:

```
exact
  (Geo.congruent_reverse_second
    A B C A).mp
    ((Geo.congruent_reverse_first
      B A C A).mp hASA.1)
```

The remaining code establishes the two noncollinearity variants and reorients the given angle congruence. These are formal interface obligations, not additional geometric constructions.

8 Relationship with Earlier Propositions

8.1 Proposition I.3

The contradiction argument assumes that one of the two sides is longer than the other and then cuts from the longer side a segment congruent to the shorter one.

This is precisely the segment-cutting operation established in Proposition I.3. It creates the auxiliary point from which the isosceles configuration used in the proof is obtained.

8.2 Proposition I.4

Once the auxiliary point has been constructed, SAS is used to compare the relevant triangles. Proposition I.4 transports the available side and angle congruences into the additional congruence information needed for the contradiction.

8.3 Proposition I.5

Proposition I.6 is the converse of Proposition I.5.

I.5 states that equal sides imply equal opposite angles:

$$AB \cong AC \implies \angle ABC \cong \angle BCA.$$

I.6 reverses the implication:

$$\angle ABC \cong \angle BCA \implies AB \cong AC.$$

The two propositions therefore establish the first clean equivalence between side symmetry and angle symmetry in a triangle.

8.4 Transition to Proposition I.7

Proposition I.7 uses this isosceles infrastructure inside a uniqueness argument. Equal distances to the endpoints of a base generate isosceles triangles, and the resulting angle congruences help exclude a second possible vertex on the same side of the base.

Thus I.6 is part of the rigidity machinery leading directly to I.7 and then to SSS in I.8.

9 Hilbert and Book Zero Components Used

9.1 Segment Trichotomy

The formal proof must turn the assumption

$$AB \not\cong AC$$

into a usable strict comparison. Segment trichotomy separates the possibilities

$$AB < AC, \quad AB \cong AC, \quad AC < AB.$$

The congruent branch contradicts the hypothesis, while the two strict branches are treated symmetrically.

9.2 Segment Layoff

In each strict branch a point is constructed on the longer side so that the cut-off segment is congruent to the shorter side. This is the formal counterpart of Euclid's application of Proposition I.3.

9.3 Isosceles Base Angles

The auxiliary equal-side configuration is converted into angle congruence by the established Hilbert isosceles theorem.

9.4 SAS and Angle Transport

SAS and angle-congruence transitivity transport the local auxiliary equalities into the orientation required to contradict the original angle hypothesis.

10 Formalization Notes

Proposition I.6 provides one of the clearest examples of the difference between a classical geometric proof and its modern formal reconstruction.

Euclid establishes the theorem by introducing an auxiliary point, constructing an additional triangle, and arguing by contradiction. The proof occupies a central place in Book I because it demonstrates how previous propositions can be combined to derive a converse statement.

Within the Hilbert framework, the same mathematical result becomes an almost immediate consequence of the previously established Angle–Side–Angle theorem. Once ASA has been derived from Hilbert’s axioms, the converse property of isosceles triangles follows directly, without auxiliary constructions or *reductio ad absurdum*.

This proposition therefore illustrates an important principle of the CGJteamLab reconstruction: as the interface theory grows, increasingly complex classical arguments collapse into short applications of high-level synthetic theorems.

11 Dependency Summary

The two proof paths are genuinely different:

Classical:

I.3 + I.4 + contradiction -> I.6

Formal:

hilbert_asa_sides -> euclid_proposition_6

The formal proof is therefore not a line-by-line rendering of Euclid’s argument. It reuses a stronger theorem already available in the Hilbert interface.

12 Conclusion

Proposition I.6 completes the converse half of Euclid’s elementary isosceles-triangle theory.

Together, I.5 and I.6 establish the equivalence between equal sides and equal opposite angles. The classical proof of I.6 is by contradiction: if the sides were unequal, the longer side could

be cut down to create an auxiliary isosceles configuration incompatible with the assumed angle congruence.

The formal reconstruction exposes one additional logical ingredient that is almost invisible on paper: from noncongruence of the two sides one must obtain a definite strict order. Segment trichotomy supplies that step and forces the proof to account for both possible orientations of the inequality.

This side-angle equivalence becomes immediate infrastructure for Proposition I.7 and, through I.7, for the SSS theorem of Proposition I.8.